

Technical Communication

BAS301 / BAS401 / BAS301H / BAS401H

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Detailed Unit-wise Study Notes with Explanations and Examples

Course Objectives

- Understand the nature and objective of Technical Communication relevant to the workplace as engineers.
- Develop an understanding of key concepts of writing, designing, and speaking.
- Utilize technical writing skills for the purposes of Technical Communication and its exposure in various dimensions.
- Build up interpersonal communication traits that will make the transition from institution to workplace smoother and help excel in jobs.
- Apply technical communication to build a personal brand and handle crisis communication.

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UNIT 1: Fundamentals of Communication and Voice Dynamics

This unit builds the foundation of the course — what communication is, why engineers need it, how it flows in organisations, what disrupts it, and the phonetic basics (sounds, tone) that shape how speech is actually produced and perceived.

1.1 Role and Purpose of Communication

Communication is the process of exchanging information, ideas, feelings, or instructions between two or more people so that both the sender and receiver arrive at a shared understanding. For an engineer, communication is not a 'soft' add-on to technical skill — it is the medium through which technical work becomes useful to others. A brilliant design that cannot be explained to a client, a manager, or a technician on the shop floor has limited practical value.

Role of Communication in the Workplace

- **Coordination:** Aligns the efforts of different people/teams so that work does not overlap or contradict.
- **Decision-making:** Managers and engineers need accurate information to make sound technical and business decisions.
- **Knowledge transfer:** Documentation, manuals, and reports carry technical knowledge from one person/generation to the next.
- **Relationship building:** Regular, honest communication builds trust between colleagues, clients, and vendors.
- **Problem solving:** Discussing a problem openly often surfaces solutions faster than working in isolation.

Purpose of Communication

- **To inform** — e.g., a status report telling stakeholders that a project is 60% complete.
- **To persuade** — e.g., a project proposal convincing management to fund new equipment.
- **To instruct** — e.g., a standard operating procedure (SOP) for operating a machine safely.
- **To motivate** — e.g., a team leader's talk before a product launch.
- **To build relationships** — e.g., informal conversations that build team cohesion.

Example: A site engineer sends a daily progress email to the project manager (informing), later presents a case for extra manpower to finish on time (persuading), and writes a safety checklist for new workers (instructing) — all in a single day, all 'technical communication'.

1.2 Types and Flow of Communication

Types of Communication

- **Verbal communication:** Use of spoken or written words — meetings, phone calls, emails, reports.
- **Non-verbal communication:** Body language, facial expressions, gestures, posture, eye contact, and tone of voice, which often convey more than the words themselves.

- **Visual communication:** Graphs, charts, diagrams, engineering drawings, and infographics that communicate data or structure at a glance.
- **Formal communication:** Follows official channels and organisational hierarchy (e.g., a memo from HR).
- **Informal communication:** Casual exchanges outside official channels, also called the 'grapevine' (e.g., colleagues discussing a rumour over coffee).

Flow of Communication in an Organisation

- **Downward:** From superiors to subordinates — orders, policies, instructions. *Example:* A plant manager emailing a new safety protocol to all shift workers.
- **Upward:** From subordinates to superiors — reports, feedback, grievances. *Example:* A junior engineer reporting a recurring machine fault to the maintenance head.
- **Horizontal (lateral):** Between people at the same level, usually across departments. *Example:* The design team coordinating with the production team on a component change.
- **Diagonal (crosswise):** Between people at different levels and different departments, bypassing the direct hierarchy. *Example:* A quality-control executive directly contacting a machine operator in another department about a defect.

1.3 Barriers to Effective Communication

A barrier is anything that distorts, blocks, or delays the intended meaning of a message. Recognising barriers is the first step to overcoming them.

- **Semantic (language) barriers:** Use of jargon, ambiguous words, or a language not fully understood by the receiver. *Example:* Using 'CAD/CAM', 'FEA' or heavy acronyms while briefing a non-technical client.
- **Physical/environmental barriers:** Noise, poor network connectivity, distance, or physical obstruction. *Example:* A video call breaking up due to poor internet during a client demo.
- **Psychological barriers:** Fear, prejudice, stress, or lack of attention that distorts how a message is sent or received. *Example:* An employee hesitating to report an error for fear of blame.
- **Organisational barriers:** Rigid hierarchy, unclear authority, or information overload. *Example:* A suggestion from a junior staff member getting lost because it must pass through five approval levels.
- **Cultural barriers:** Differences in norms, gestures, or etiquette across regions. *Example:* A thumbs-up gesture is positive in most cultures but offensive in some parts of West Africa and the Middle East.
- **Perceptual barriers:** Two people interpreting the same message differently based on their own experience or bias.

Overcoming Barriers

- Use simple, precise, and unambiguous language suited to the audience.
- Choose the right channel (face-to-face for sensitive issues, email for records).
- Practice active listening and seek feedback to confirm understanding.
- Be culturally sensitive and aware of non-verbal cues.

1.4 The 7 C's of Communication

The 7 C's are a checklist for drafting any effective message — spoken or written.

C	Meaning	Example
Clarity	The message should be plain and easily understood, with one clear idea per sentence.	'Submit the report by 5 PM Friday' instead of 'Please try to get the report done soon.'
Conciseness	Say what is necessary in the fewest possible words, avoiding repetition.	'We are unable to proceed' instead of 'It is important to note that we are not in a position to proceed at this juncture.'
Concreteness	Use specific facts and figures rather than vague statements.	'Sales rose by 12% in Q2' instead of 'Sales improved a good deal.'
Correctness	Accurate grammar, facts, figures, and appropriate level of language.	Double-checking a technical specification before sending it to a client.
Coherence	Ideas should be logically connected and flow in a sensible order.	A report that moves from Introduction to Method to Results to Conclusion, not jumbled.
Completeness	The message must include all the facts the reader needs to act on it.	A meeting invite that states date, time, venue, and agenda — not just 'let's meet.'
Courtesy	Respectful, friendly, and considerate tone, even while refusing or correcting.	'Thank you for your query; unfortunately we cannot meet this deadline' instead of 'That's not possible.'

1.5 Code and Content

Every message has two components. The **content** is the actual idea, fact, or feeling being communicated — the 'what'. The **code** is the system of symbols (a language, gestures, mathematical notation, an engineering drawing convention) used to represent that content — the 'how'. For communication to succeed, sender and receiver must share the same code, or a translation/interpretation step is required.

***Example:** An engineering drawing uses standard symbols (the code) to represent dimensions and tolerances (the content). If a foreign vendor is unfamiliar with a national drafting standard, the content — however correct — fails to transfer, because the code is not shared.*

Encoding is the sender converting an idea into a code (words, symbols); decoding is the receiver interpreting that code back into meaning. Miscommunication frequently arises not because the content was wrong, but because the code was unfamiliar, ambiguous, or misused (e.g., using an abbreviation the other person has never seen).

1.6 Stimulus and Response

Communication can be modelled as a stimulus-response cycle, borrowed from behavioural psychology. A **stimulus** is anything that triggers a reaction — a spoken word, a written notice, a gesture, an alarm. The **response** is the receiver's reaction to that stimulus, which itself becomes a new stimulus for the original sender, creating a continuous loop of interaction (this is why communication is described as a two-way, ongoing process, not a one-time event).

Example: A supervisor blows a whistle (stimulus) to signal a shift change; workers stop their machines (response). That visible response is itself a new stimulus, confirming to the supervisor that the message was received and understood.

Understanding this cycle helps a communicator predict how an audience is likely to react and adjust the message (tone, channel, timing) to get the intended response, rather than an unintended one such as confusion, offence, or inaction.

1.7 Voice Dynamics: Vowel Sounds, Consonant Sounds, and Tone

This part of the unit deals with the mechanics of spoken English — how sounds are produced — which underlies clear pronunciation and effective oral presentation.

Vowel Sounds

Vowels are speech sounds produced with an open vocal tract — no blockage of airflow by the tongue, teeth, or lips. In English there are traditionally 12 pure vowel sounds (monophthongs, e.g., the sounds in 'bit', 'bat', 'boot') and 8 diphthongs (glides between two vowel sounds within one syllable, e.g., the sound in 'coin' or 'now'). Vowels form the 'core' or nucleus of every syllable and carry much of the audible energy and pitch of speech.

Example: The word 'seat' has a single long vowel sound; the word 'coin' has a diphthong that glides from one vowel quality to another within the same syllable.

Consonant Sounds

Consonants are produced with some obstruction or narrowing of the airflow using the lips, teeth, tongue, or palate. English has 24 consonant sounds, classified by (a) place of articulation — where the obstruction occurs (lips, teeth, palate, throat), and (b) manner of articulation — how the air is obstructed (stops/plosives like /p/, /b/; fricatives like /f/, /s/; nasals like /m/, /n/; and so on).

Example: The sounds /p/ and /b/ are both 'bilabial plosives' — made by briefly closing both lips and releasing a burst of air — but /p/ is voiceless (no vocal cord vibration) while /b/ is voiced.

Tone: Rising and Falling Tone

Tone (or intonation) is the rise and fall of pitch in speech, and it changes the meaning or implication of a sentence even when the words stay the same.

- **Falling tone** (pitch drops toward the end): used for statements of fact, commands, and 'wh-' questions, signalling certainty or finality.
- **Rising tone** (pitch rises toward the end): used for yes/no questions, to express doubt, surprise, or to invite the listener to continue, signalling incompleteness or a request for a response.

Example: 'You finished the report.' spoken with a falling tone is a statement confirming a fact. The exact same words spoken with a rising tone — 'You finished the report?' — become a surprised or doubting question.

For engineers, controlling tone matters in situations like presenting results confidently (falling tone on key conclusions) versus checking a client's understanding (rising tone at the end of a confirming question, e.g., 'Does that specification work for you?').

UNIT 2: Communication Skills for Career Building

This unit is directly aimed at employability — the documents and spoken skills students need to secure a job, pursue higher studies, and represent themselves and their institution at professional forums.

2.1 CV and Resume Writing

A **resume** is a brief (usually 1-2 page) summary of education, skills, and experience tailored for a specific job. A **Curriculum Vitae (CV)** is a longer, more comprehensive document listing full academic and professional history — publications, research, conferences — commonly used for academic, research, or international applications.

Key Sections

- **Contact information:** Name, phone, professional email, LinkedIn/portfolio link.
- **Career objective/summary:** 2-3 lines stating the role sought and the value offered.
- **Education:** Degree, institution, year, CGPA — in reverse chronological order.
- **Technical skills:** Programming languages, software, tools relevant to the role.
- **Projects/Internships:** Title, brief description, your specific contribution, and outcome.
- **Achievements/Certifications:** Awards, competitions, relevant certifications.
- **Extra-curricular activities:** Leadership or teamwork evidence (clubs, sports, volunteering).

Writing Principles

- Use action verbs: 'Designed', 'Implemented', 'Led', 'Optimised' — not passive phrases like 'Was responsible for'.
- Quantify achievements wherever possible.
- Keep formatting consistent, and tailor the resume to each job description.

***Example:** Weak: 'Worked on a project about traffic management.' Strong: 'Designed an IoT-based traffic signal controller that reduced simulated intersection wait time by 18%, using Arduino and Python.'*

2.2 Interview Skills

An interview is a structured conversation to assess a candidate's suitability for a role, combining technical evaluation with communication, attitude, and cultural fit.

Before the Interview

- Research the company, role, and recent projects/news.
- Revise your resume thoroughly — you can be asked about anything written on it.
- Prepare answers for common questions ('Tell me about yourself', 'Why should we hire you?').
- Prepare 2-3 thoughtful questions to ask the interviewer.

During the Interview

- Maintain good posture, eye contact, and a firm handshake (in-person) or a professional setup and eye-level camera (virtual).
- Use the STAR method for behavioural questions: Situation, Task, Action, Result.
- Listen fully to the question before answering; ask for clarification if needed.
- Be honest about what you don't know, but show willingness to learn.

Example: Q: 'Tell me about a time you handled conflict in a team.' Using STAR — Situation: two teammates disagreed on the design approach for a mini-project. Task: as team lead, I needed a decision without demoralising anyone. Action: I organised a quick pros/cons comparison and let data decide. Result: the team adopted the more efficient design and finished two days early.

2.3 Group Discussion (GD)

A Group Discussion is a selection technique where a group of candidates (usually 6-10) discusses a topic for 10-15 minutes so evaluators can assess communication, subject knowledge, leadership, and teamwork simultaneously.

Skills Evaluated

- Clarity of thought and ability to structure an argument.
- Listening and building on others' points rather than repeating them.
- Assertiveness without being aggressive; ability to disagree respectfully.
- Leadership shown by summarising, initiating, or steering the discussion back on track.

Do's and Don'ts

- Do: initiate confidently if you have a strong opening point; use data/examples to support views.
- Do: bring in silent members' opinions occasionally — it signals leadership.
- Don't: interrupt others or dominate the entire discussion.
- Don't: stay silent throughout — silence is scored as poorly as over-aggression.

Example: Topic: 'Should engineering colleges make internships compulsory?' A good participant might open with, 'I feel internships should be compulsory because they close the gap between theoretical learning and industry expectation — for instance, most recruiters today rate practical exposure above CGPA.'

2.4 Effective Profiling

Profiling refers to building an accurate, appealing, and consistent professional identity across all platforms and documents — resume, LinkedIn, portfolio, and how you present yourself verbally. Effective profiling means your strengths are clearly identifiable and consistently represented, so recruiters and collaborators quickly understand 'what you are good at' and 'what you stand for' professionally.

Example: An engineering student aiming for a career in embedded systems keeps their LinkedIn headline, resume objective, GitHub projects, and elevator pitch all consistently focused on embedded/IoT work, rather than presenting themselves as a generalist in every document.

2.5 Communication and Networking for Building Relationships

Networking is the deliberate building and maintaining of professional relationships that can provide information, referrals, mentorship, or opportunities over time. Good networking is based on genuine, two-way communication rather than one-sided requests.

- **In-person networking:** Conferences, seminars, alumni meets — introduce yourself with a short, clear pitch, and follow up afterward.
- **Online networking:** LinkedIn connections, professional forums, contributing to open-source projects or technical communities.
- **Maintaining relationships:** Periodic, non-transactional check-ins; congratulating contacts on achievements; offering help before asking for it.

Example: After a technical seminar, following up with a speaker via a short LinkedIn message referencing a specific point from their talk is far more effective networking than a generic 'please connect' request.

2.6 Writing the Statement of Purpose (SOP)

An SOP is an essay submitted with applications for higher studies (MS/MTech/PhD abroad or in India) explaining who you are academically, why you want to pursue this specific programme, and how it fits your long-term goals.

Typical Structure

- **Opening hook:** A specific incident/motivation that sparked your interest in the field.
- **Academic background:** Relevant coursework, projects, and research that built your foundation.
- **Professional/research experience:** Internships, publications, and the specific skills gained.
- **Why this programme/university:** Specific faculty, labs, or courses you want to work with — shows genuine research rather than a generic template.
- **Future goals:** How this degree connects to your career plan.

Example: Instead of 'I have always been passionate about robotics', a stronger opening might be: 'While building a line-following robot in my second year, I spent three sleepless nights debugging a PID controller — and that frustration turned into the question that still drives me: how do we make control systems more intuitively tunable?'

2.7 Seminar and Conference Paper Writing

A seminar/conference paper is a structured written report of research or study, submitted for presentation or publication, following a formal academic format.

Standard Sections (IMRaD format)

- **Title and Abstract:** A concise (150-250 word) summary of the problem, method, and key findings.
- **Introduction:** Background, problem statement, and objective of the work.
- **Literature Review:** Summary of prior related work and the gap your work addresses.
- **Methodology:** How the study/experiment was conducted.
- **Results and Discussion:** Data, findings, and their interpretation.
- **Conclusion and Future Scope:** Summary of contribution and possible extensions.
- **References:** Properly cited sources, in the required citation style (IEEE, APA, etc.).

Example: A paper titled 'Low-Cost IoT-Based Soil Moisture Monitoring for Precision Irrigation' would follow this structure, with the abstract stating the sensor used, the accuracy achieved, and the cost saving versus commercial systems.

2.8 Expert Technical Lecture: Writing and Presenting

A technical lecture communicates specialised knowledge to an audience that may range from peers to laypersons, and requires careful planning of both content and delivery.

Writing the Lecture

- Define the audience's existing knowledge level and tailor depth accordingly.
- Organise content logically: context → core concept → application → summary.
- Support abstract ideas with analogies, diagrams, or real-world case studies.

Presenting the Lecture

- Open with a hook — a question, statistic, or short story that creates interest.
- Use visual aids (slides, diagrams) to support, not duplicate, spoken words.
- Pace delivery, pause for emphasis, and invite questions at natural breaks.
- Close with a clear takeaway message the audience should remember.

Example: A lecture on 'Introduction to Machine Learning for Mechanical Engineers' might open with a relatable analogy — comparing a trained ML model to a fitted engineering curve — before introducing formal terms like features, labels, and loss functions.

UNIT 3: Communication Skills for Presentation: Writing, Designing, and Speaking

This unit covers the major long-form technical documents engineers must produce (thesis, project reports, proposals) and the skills needed to pitch ideas and deliver spoken presentations effectively.

3.1 Thesis and Project Report Writing

A thesis or project report is a formal, detailed written account of a research or design project, demonstrating the problem addressed, the method used, and the results achieved.

Typical Structure

- **Title page and Certificate:** Formal declaration and approval by guide/institution.
- **Abstract:** One-page overview of the entire work.
- **Introduction:** Background, problem statement, objectives, and scope.
- **Literature Survey:** Review of existing solutions/research and identified gaps.
- **Methodology/Design:** Detailed explanation of the approach, tools, and techniques used.
- **Implementation and Results:** What was built/tested and the outcomes, with data, graphs, and tables.
- **Conclusion and Future Work:** Summary of achievement and possible improvements.
- **References and Appendices:** Cited sources and supplementary material (code, datasheets).

Writing Guidelines

- Use formal, objective, third-person language ('The system was tested...' not 'I tested the system...', unless the institute's format specifically allows first person).
- Number all figures and tables, and refer to them in the text (e.g., 'as shown in Figure 3.2').
- Maintain consistency in units, terminology, and formatting throughout.

Example: Rather than writing 'We made the circuit and it worked well', a thesis would state: 'The proposed circuit was implemented on a PCB and tested under three load conditions; it maintained output ripple below 2%, meeting the design specification.'

3.2 Technical Proposal Writing

A technical proposal is a persuasive document written to convince a client, funding body, or management to approve a project, allocate a budget, or accept a solution.

Key Components

- **Problem statement:** The need or gap the proposal addresses.
- **Proposed solution:** Technical approach, with enough detail to establish credibility without overwhelming the reader.
- **Timeline:** Milestones and delivery schedule, often shown as a Gantt chart.
- **Budget/Resources:** Cost breakdown and resources required.

- **Benefits/Justification:** Why this solution is the best choice, addressing feasibility and ROI.

Example: A proposal to a factory manager for replacing manual inspection with a machine-vision defect detector would state the current defect-escape rate, propose the camera-based system, estimate cost versus manual-labour savings over two years, and give an implementation timeline.

3.3 How to Pitch an Idea: Process, Preparation, and Structure

Pitching is a short, high-impact form of persuasive communication used to sell an idea, product, or project — often to investors, management, or a panel — usually within a strict time limit.

Process

- **Research:** Understand the audience's priorities and constraints before pitching.
- **Distill:** Reduce the idea to its single most compelling value proposition.
- **Rehearse:** Practice for timing, clarity, and handling likely questions.

Structure (a common framework)

- **Hook:** Open with a striking problem statement or statistic.
- **Problem:** Explain who is affected and why it matters.
- **Solution:** Present your idea/product as the answer, ideally with a quick demo.
- **Value/Differentiation:** Why this solution over existing alternatives.
- **Ask:** The specific action you want from the audience (funding, approval, partnership).

Example: An elevator pitch for a campus project: 'Every year, over 200 students in our hostel miss laundry slots because bookings are managed on a physical notice board. We built a simple app that lets students book and track washing-machine slots in real time, cutting average wait time by 40%. We're asking the hostel administration for permission to pilot it in Block C next semester.'

3.4 Elements of Speech Delivery: Passion, Poise, and Illustrations

Beyond content, how a speech is delivered strongly determines its impact.

Passion

Genuine enthusiasm for the subject is contagious — it is conveyed through vocal energy, varied pitch, and animated (but controlled) body language. An audience senses when a speaker actually cares about what they are saying, and this significantly increases engagement and credibility.

Poise

Poise is composure and confidence under pressure — steady posture, controlled pace, and calm recovery from mistakes (e.g., not panicking if a slide fails to load, but smoothly continuing). Poise is built through preparation, practice, and familiarity with the material.

Illustrations

Illustrations — stories, analogies, case studies, or visual aids — make abstract technical points concrete and memorable. A well-chosen illustration can explain in ten seconds what might take several sentences of pure explanation.

Example: Explaining 'redundancy in engineering systems' abstractly can be dry. Illustrating it with 'this is why aircraft have multiple independent hydraulic systems — if one fails, the others keep the plane controllable' makes the concept instantly graspable and memorable.

UNIT 4: Communication and Leadership Development

This unit connects communication to leadership and interpersonal effectiveness — how leaders communicate, how feedback and trust are built, and the thinking skills that underlie sound communication and decision-making.

4.1 Leadership Communication

Leadership communication is the way a leader conveys vision, expectations, and decisions to influence and align a team toward shared goals. Unlike routine managerial communication, it carries a persuasive and motivational dimension — leaders communicate not just facts but purpose.

Characteristics of Effective Leadership Communication

- **Vision-oriented:** Connects daily tasks to a larger goal ('we're not just writing code, we're building the system that will run the plant').
- **Transparent:** Shares information honestly, including bad news, to maintain credibility.
- **Two-way:** Invites questions and dissent rather than only issuing directives.
- **Consistent:** Words match actions over time, which builds long-term trust.

***Example:** Before a tight project deadline, a leader who says, 'I know this timeline is aggressive; here is exactly why it matters to the client, and here is how I plan to support you if you get stuck' communicates leadership far more effectively than one who simply says 'Just get it done.'*

4.2 Communication and Social Competence: Context, Feelings, Intentions, Behaviours

Social competence is the ability to read a social situation accurately and respond appropriately — a critical skill for effective interpersonal communication.

- **Context:** Recognising the setting (formal meeting vs. casual chat) and adjusting language, tone, and formality accordingly.
- **Feelings:** Being aware of one's own emotional state and reading the emotional state of others (e.g., noticing a colleague seems stressed before delivering critical feedback).
- **Intentions:** Understanding the underlying goal behind a message — is the speaker seeking advice, venting, or making a request? — and responding to the real intention rather than only the literal words.
- **Behaviours:** The observable actions (tone, word choice, body language) through which context, feelings, and intentions are actually expressed and perceived by others.

***Example:** A colleague says, 'This deadline is impossible.' If read only literally, one might argue about feasibility. Reading the underlying feeling (stress) and intention (seeking support, not a debate) suggests a better response: 'That does sound tight — what part is most stuck? Let's see if I can help.'*

4.3 Providing and Receiving Feedback

Feedback is information given about performance or behaviour, intended to reinforce strengths or correct gaps. It is central to learning and growth in any professional setting.

Giving Effective Feedback

- Be specific and behaviour-focused, not personal ('The report was missing the cost analysis section' rather than 'You're careless').
- Balance positive and constructive points — the widely taught 'sandwich' method places constructive feedback between two genuine positive observations.
- Give feedback promptly and privately when it is corrective.
- Focus on actionable next steps, not just what went wrong.

Receiving Feedback

- Listen without interrupting or becoming defensive.
- Ask clarifying questions if the feedback is vague.
- Separate the feedback about the work from your self-worth.
- Thank the giver — it encourages a continued culture of honest feedback.

***Example:** Instead of 'Your presentation was bad', constructive feedback would be: 'Your data analysis was thorough (positive) — the slides had a lot of text that made it hard to follow live (constructive); try cutting each slide to one key idea with a supporting chart (actionable), and your explanations were clear otherwise (positive).'*

4.4 Difference between Tact and Intelligence

Intelligence is the cognitive capacity to acquire and apply knowledge — to reason, analyse, and solve problems. **Tact** is the skill of communicating truth or difficult information in a way that is sensitive to the other person's feelings and the social situation. A person can be highly intelligent yet lack tact (blunt, technically correct but socially damaging), or be tactful without being particularly analytically sharp.

***Example:** An intelligent-but-tactless response to a colleague's flawed design: 'This design won't work; the load calculations are wrong.' A tactful-and-equally-intelligent response: 'The overall concept is solid — I think the load calculation for the beam section might need a second look; want to go through it together?' Both identify the same technical error, but the second preserves the relationship and is more likely to be well received.*

4.5 Emotional Intelligence (EI)

Emotional Intelligence is the ability to recognise, understand, and manage one's own emotions, and to recognise and influence the emotions of others. Daniel Goleman's widely used model identifies five components.

- **Self-awareness:** Recognising one's own emotions and their effect on thoughts and behaviour.
- **Self-regulation:** Managing disruptive emotions and impulses.
- **Motivation:** Being driven to achieve for internal reasons beyond money or status.
- **Empathy:** Understanding others' emotional makeup and responding appropriately.
- **Social skills:** Managing relationships to move people in desired directions.

Example: An engineer with high EI, upon realising a design flaw was caused by a team member's mistake, manages their own frustration (self-regulation), considers how the teammate must be feeling (empathy), and raises the issue constructively in private rather than calling it out in a group meeting (social skill).

4.6 Trust through Communication

Trust is the confidence that another person's word and actions are reliable. Communication is the primary vehicle through which trust is built or eroded over time.

- **Consistency:** Doing what you say you will do, repeatedly.
- **Transparency:** Sharing relevant information openly, including uncomfortable truths.
- **Reliability in small commitments:** Trust is built more through many small honoured promises than through occasional grand gestures.
- **Active listening:** Making others feel heard increases their trust in the relationship.

Example: A project lead who consistently gives realistic (rather than overly optimistic) timelines, and flags risks early rather than hiding them until they become crises, builds far more long-term trust with stakeholders than one who always promises the best-case scenario.

4.7 Thinking Skills: Meaning and Types

Thinking skills are the mental processes used to process information, solve problems, and make decisions — they underpin how clearly and logically a person communicates.

- **Critical thinking:** Objectively analysing and evaluating information to form a judgement — questioning assumptions and evidence rather than accepting claims at face value.
- **Creative thinking:** Generating novel ideas or unconventional solutions to a problem.
- **Analytical thinking:** Breaking a complex problem into smaller components to understand it systematically.
- **Logical thinking:** Reasoning step-by-step from premises to a valid conclusion.
- **Lateral thinking:** Approaching a problem from an unexpected angle rather than the obvious, direct route (a term coined by Edward de Bono).
- **Reflective thinking:** Reviewing one's own experience or reasoning to learn from it.

Example: Faced with a bridge design that keeps exceeding budget: analytical thinking breaks the cost into materials, labour, and design components; critical thinking questions whether the original material specification is actually necessary; lateral thinking might suggest an entirely different bridge type (e.g., a cable-stayed instead of a truss design) that the team hadn't considered.

UNIT 5: Digital Communication and Personality Making

This unit covers communicating through digital/social media channels responsibly and effectively, and closes the course with the psychology of personality — since how one communicates is deeply shaped by personality.

5.1 Content Creation for Social Media: Emails, Webinars, Podcasts, Blogs

Professional Emails

Professional email remains the backbone of formal digital communication. A good email has a clear, specific subject line, a courteous greeting, a concise body (ideally answering 'what, why, and what action is needed' within the first few lines), and a professional sign-off.

Example: Subject: 'Action needed: Approval for Component Change – Due Friday' is far more effective than a vague subject like 'Update'.

Webinars

A webinar is a live, online seminar. Effective webinar communication requires clear visual slides, engaging the (unseen) audience through direct address and interactive polls/Q&A, and managing technical aspects (audio/video quality, screen-sharing) smoothly since technical glitches disproportionately damage credibility online.

Podcasts

A podcast is an audio-only or audio-first content series. Since there is no visual aid, clarity of speech, vocal variety, structured segments, and strong verbal storytelling carry the entire communicative load.

Blogs

A blog is a written, informal-to-semi-formal online article, usually structured with a catchy headline, short paragraphs, subheadings for scannability, and a conversational but credible tone — balancing personality with accuracy.

Example: A technical blog post explaining a new sensor might open with a relatable problem ('Ever wondered how your phone knows which way is up?'), then explain the accelerometer in simple terms with a diagram, before going into technical depth for interested readers.

5.2 Effective and Ethical Use of Social Media

Social media dramatically amplifies both the reach and the risk of communication, making ethical use essential.

Effective Use

- Know your platform: LinkedIn for professional content, Twitter/X for concise updates, Instagram/YouTube for visual storytelling — tailor tone and format accordingly.
- Maintain consistency in posting and personal branding.

- Engage genuinely with comments rather than only broadcasting.

Ethical Use

- Verify facts before sharing/forwarding — avoid spreading misinformation.
- Respect privacy: do not share others' personal information or images without consent.
- Give credit for others' work; avoid plagiarism.
- Maintain professionalism — remember that online posts are often public and permanent, and can affect your professional reputation ('digital footprint').

Example: Before resharing a viral claim about an engineering failure, an ethical communicator checks the original source rather than immediately reposting an unverified claim that could damage a company's or individual's reputation.

5.3 Text and Technique

'Text' here refers to the actual written content, and 'technique' to the craft used to shape it for its medium and purpose — word choice, sentence length, formatting, and use of visuals, all adapted to whether the text will be read on a phone screen, in a formal report, or heard aloud.

Example: The same product update might be written as a two-line tweet ('New firmware v2.1 fixes the Bluetooth pairing bug — update now!'), a detailed changelog for a technical audience, and a longer blog post for general users — same core content, different text technique for each medium.

5.4 Speech and Personality

The way a person speaks — vocabulary, pace, tone, confidence, humour — is one of the most immediate windows into their personality, and consciously shaping one's speech is part of shaping one's professional persona ('personality making').

Example: A naturally introverted engineer preparing for client-facing roles might deliberately practise more expressive vocal variety and structured small talk — not to change who they are, but to communicate their existing expertise more accessibly.

5.5 Personality Analysis: Types of Personality

Several established frameworks classify personality types; understanding them helps in self-awareness and in adapting communication to different personality types.

Big Five (OCEAN) Model — a widely used trait-based model

- **Openness:** Curiosity and willingness to try new experiences/ideas.
- **Conscientiousness:** Organisation, discipline, and reliability.
- **Extraversion:** Sociability and energy drawn from external interaction.
- **Agreeableness:** Cooperativeness and consideration for others.
- **Neuroticism:** Tendency toward emotional instability versus calmness.

Introvert–Extrovert–Ambivert (a simpler, common classification)

- **Introverts:** Gain energy from solitude; often prefer written or one-on-one communication.
- **Extroverts:** Gain energy from social interaction; often thrive in group discussion and public speaking.

- **Ambiverts:** Show a balance of both traits depending on the situation.

***Example:** An introverted team member may contribute more effectively in written form (detailed emails, documentation) while an extroverted teammate may be the natural choice to lead a client-facing verbal presentation — good team communication design plays to these strengths.*

5.6 Concept of Personality: Maslow, Freud, Vivekananda, Jung Typology & Personality Assessment

Maslow — Hierarchy of Needs

Abraham Maslow proposed that human motivation follows a hierarchy: Physiological needs (food, water) → Safety → Love/belonging → Esteem → Self-actualisation (realising one's full potential). Maslow argued personality development is closely tied to how well these needs are met, with self-actualisation representing the highest, most complete stage of personal growth.

***Example:** An employee worried about job security (a safety need) is unlikely to be motivated by appeals to 'creative self-expression' (a self-actualisation need) until the safety concern is addressed first.*

Freud — Psychoanalytic Theory

Sigmund Freud proposed personality is shaped by three interacting components: the **Id** (primitive, pleasure-seeking instincts), the **Ego** (the rational part that mediates between the Id and reality), and the **Superego** (internalised moral standards and conscience). Freud believed much of personality and behaviour is influenced by unconscious processes and early childhood experiences.

***Example:** A person tempted to take credit for a colleague's idea (the Id's self-interest) but who instead acknowledges the colleague publicly (guided by the Superego's sense of fairness, balanced by the Ego's practical judgement about maintaining the relationship).*

Vivekananda — Indian Philosophical Perspective

Swami Vivekananda's view of personality emphasises the development of inner strength, self-confidence, and character through self-discipline and service to others, drawing on Vedanta philosophy. He emphasised that true personality development comes from realising one's inner potential ('each soul is potentially divine') and channelling it through fearlessness, focus, and selfless action, rather than purely external achievement.

Jung — Typology

Carl Jung proposed that individuals have inborn preferences along key dimensions of psychological function, which forms the theoretical basis for later tools such as the Myers-Briggs Type Indicator (MBTI). Key Jungian dimensions include:

- **Extraversion vs. Introversion:** Orientation of psychic energy — outward to the external world, or inward to one's own thoughts.
- **Sensing vs. Intuition:** How information is perceived — through concrete facts and senses, or through patterns and possibilities.
- **Thinking vs. Feeling:** How decisions are made — through logical analysis, or through values and impact on people.

Personality Assessment

Personality assessment refers to structured tools used to measure personality traits, commonly used in recruitment, career counselling, and self-development.

- **MBTI (Myers-Briggs Type Indicator):** Classifies individuals into 16 types based on Jung's dimensions (e.g., INTJ, ESFP).
- **Big Five/OCEAN inventory:** A trait-scored questionnaire measuring the five factors described earlier.
- **Projective tests** (e.g., Thematic Apperception Test): Present ambiguous stimuli and infer personality from the individual's interpretation.

***Example:** A company might use an MBTI-style assessment during recruitment to understand whether a candidate's natural working style (e.g., detail-oriented and structured, or big-picture and flexible) fits a particular team's way of working — not to select 'better' or 'worse' candidates, but to understand communication style and team fit.*

Prescribed Books

1. Technical Communication – Principles and Practices by Meenakshi Raman & Sangeeta Sharma, Oxford University Press, 2018, New Delhi.
2. Personality Development and Soft Skills by Barun K. Mitra, OUP, 2012, New Delhi.
3. Technical Communication by Pfeiffer, 6th edn (Pearson).
4. Soft Skills & Employability by Sabina Pillai and Agna Fernandez, Cambridge University Press, 2018.
5. Practical Communication: Process and Practice by L.U.B. Pandey, A.I.T.B.S. Publications India Ltd., Krishan Nagar, 2014, Delhi.